Table 15.1 Types of circuit breakers and their memory settings

Circuit breaker	Description and minimum memory limit	Property
Parent	The total memory that can be used across all the other circuit breakers. Defaults to 70% of JVM heap memory if real-time memory is taken into account (`indices.breaker.total.use_real_memory=true`); otherwise, it is 95% of JVM heap memory.	`indices.breaker.total.limit`
Inflight requests	The sum total of memory of all inflight requests, which is not to be exceeded by a threshold. The default is 100% of JVM heap memory, although it derives the actual percentage from the parent circuit breaker.	`network.breaker.inflight_requests.limit`
Request	Helps prevent exceeding the heap memory to serve a single request. The default is 60% of the JVM heap.	`indices.breaker.request.limit`
Field data	Helps prevent exceeding memory when loading fields into the `fields` cache. The default limit is 40% of heap memory.	`indices.breaker.fielddata.limit`
Accounting requests	Avoids accumulating memory after the request is served. The default is 100%, which means it inherits the threshold from the parent.	`indices.breaker.accounting.limit`
Script compilation	While all other circuit breakers look after memory, this circuit breaker limits the number of inline compilations in a fixed time period. The default is 150/5 min (150 script compilations in a period of 5 minutes).	`script.max_compilations_rate`

Circuit breakers help avoid using excess memory for few operations that occur frequently. This helps to maintain a stable cluster.

15.5 Final words

Because a huge number of things can go wrong on a production cluster, it is not only impractical but impossible to cover all performance and troubleshooting problems in this chapter. Those we have looked at barely scratch the surface. Most problems require a detailed investigation, profiling of the application, sifting through logs, trial-and-erroring a few options, and so on. A word of advice: stay calm and be methodical in your journey of maintaining a healthy and stable cluster.

I also advise you to take your Elasticsearch cluster for a drive so you understand it thoroughly inside and out. Experimenting with larger (future-proof) data sets in a